MAT313: Differential Geometry for Curves and Surfaces

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1 Lecture 01

§1.1 Background discussion

Convention. A map is *differentiable* if it has continuous derivatives of all orders at all points of its domain.

A **parametrized differentiable curve** is a differentiable map

$$\alpha : I \to \mathbb{R}^3 \tag{1.1}$$

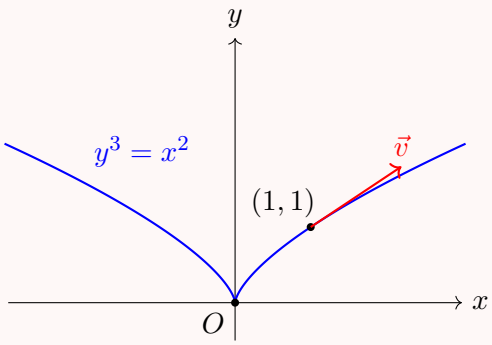
of an open interval[1] $I = (a, b)$ of $\mathbb{R}$ into $\mathbb{R}^3$.

Example 1.1.1

The map

$$\alpha : \mathbb{R} \to \mathbb{R}^3 \quad \text{by} \quad \alpha(t) = (t^3, t^2), \quad \text{where} \quad t \in \mathbb{R} \tag{1.2}$$

is a parametrized differentiable curve. Here, $x = t^3$ and $y = t^2$. Eliminate t to obtain $y^3 = x^2$. Therefore, the curve has the following trace:



In this example, $\alpha'(t) = (3t^2, 2t)$. At $t = t_0 \neq 0$, the slope of the tangent line is $m = \frac{2t_0}{3t_0^2} = \frac{2}{3t_0}$. And the tangent line goes through $\alpha(t_0) = (t_0^3, t_0^2)$. Hence, the equation of the tangent at $t = t_0$ is

$$y - t_0^2 = \frac{2}{3t_0}\left(x - t_0^3\right). \tag{1.3}$$

Definition 1.1.2. Let $\alpha : I \to \mathbb{R}^3$ be a parametrized differentiable curve. For each $t \in I$ where $\alpha'(t) \neq 0$, there is a well-defined straight line which goes through $\alpha(t)$ and contains the vector $\alpha'(t)$. This line is called the **tangent line to** α **at** t. Any point t where $\alpha'(t) = 0$ is a **singular point** of α.

In Example 1.1.1, $t = 0$ is a singular point. We restrict our attention to curves without any singular points.

Definition 1.1.3 (Regular curve)**.** A parametrized differentiable curve $\alpha : I \to \mathbb{R}^3$ is said to be **regular** if $\alpha'(t) \neq 0$ for all $t \in I$.

[1] The word *interval* is taken in a generalised sense, so that we do not exclude the cases $a = -\infty$, $b = \infty$.

Given $t \in I$, the arc length of a regular parametrized curve $\alpha : I \to \mathbb{R}^3$ from the point t_0 is

$$s(t) = \int_{t_0}^{t} \|\alpha'(u)\| du, \tag{1.4}$$

where

$$\|\alpha'(u)\| = \sqrt{(x'(u))^2 + (y'(u))^2 + (z'(u))^2} \tag{1.5}$$

given that $\alpha(u) = (x(u), y(u), z(u))$, and so $\alpha'(u) = (x'(u), y'(u), z'(u))$.

The arc length $s(t)$ is a differentiable function of t on the domain I. Indeed, $\|\alpha'\| = \|\cdot\| \circ \alpha'$ is continuous, for being the composition of two continuous functions. Therefore, by the First Fundamental Theorem of Calculus,

$$s(t) = \int_{t_0}^{t} \|\alpha'(u)\| du \tag{1.6}$$

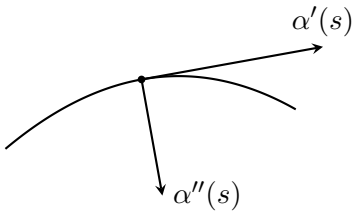
$$\implies s'(t) = \|\alpha'(t)\|. \tag{1.7}$$

It may well be the case that the parameter t itself is the arch length s, i.e., $s = t$. Then $\frac{ds}{dt} = 1 = \|\alpha'(t)\|$. And so

$$s = \int_{t_0}^{t} \|\alpha'(u)\| du = \int_{t_0}^{t} 1 du = t - t_0. \tag{1.8}$$

§1.2 The local theory of curves parametrized by arc length

Let $\alpha : I = (a, b) \to \mathbb{R}^3$ be a curve parametrized by arc length s. We know that the speed is $\|\alpha'(s)\| = 1$. Hence, the acceleration is normal[2] to the tangent vector at the point. This acceleration vector $\alpha''(s)$ pulls the object and deviates it from the tangent line. If it were zero, the object would have travelled along the tangent line. The more the acceleration, i.e., the more $\|\alpha''(s)\|$ is, the more the deviation of the trajectory of the object or curve from the tangent line at s.



Definition 1.2.1 (Curvature)**.** Let $\alpha : I \to \mathbb{R}^3$ be a curve parametrized by arc length $s \in I$. The number

$$\kappa(s) \coloneqq \|\alpha''(s)\| \tag{1.9}$$

is called the **curvature of** α at s.

Orientation a la do Carmo. The ambient space where the curves and surfaces are living in is $\mathbb{R}^3$. The curve $\alpha(s)$ living in $\mathbb{R}^3$ has a velocity vector

$$\alpha'(s) = \begin{bmatrix} x'(s) \\ y'(x) \\ z'(s) \end{bmatrix}. \tag{1.10}$$

[2] $\alpha(s) \cdot \alpha(s) = 1 \implies 2\alpha''(s) \cdot \alpha'(s) = 0 \implies \alpha''(s) \perp \alpha'(s)$

To discuss the orientation of this curve, do Carmo reverses the direction of traversal as the parameter of the curve increase.

Recall the curvature $\kappa(s)$ of α at s, given by Equation 1.9. At points where $\kappa(s) \neq 0$, a unit vector $n(s)$ in the direction of $\alpha'(s)$ is well-defined by the equation

$$\alpha''(s) = \kappa(s) \cdot n(s). \tag{1.11}$$

From $\alpha'(s) \cdot \alpha'(s) = 1$, one obtains by differentiation $\alpha''(s) \cdot \alpha'(s) = 0$, and from Equation 1.11 one finds that $n(s)$ is normal to $\alpha'(s)$. Indeed,

$$n(s) \cdot \alpha'(s) = \frac{\alpha''(s)}{\kappa(s)} \alpha'(s) = 0. \tag{1.12}$$

$n(s)$ is called the **normal vector at** s.

The plane determined by the unit tangent vector $\alpha'(s)$ and the normal vector $n(s)$ is called the **osculating plane**.

Definition 1.2.2 (Singular points of order 0 and 1)**.** For the arc-parametrized curve $\alpha(s)$, s is a **singular point of order** 0 if $\alpha'(s) = 0$ and it is a **singular point of order** 1 if $\alpha''(s) = 0$ with $\alpha'(s) \neq 0$.

Example 1.2.3 (Planar inflection curve)

The curve $\alpha(t) = (t, t^3, 0)$ is an inflection curve lying in the xy-plane. Note that it's not arc-parametrized. In fact, to understand singular points of order 0 and 1, one doesn't need an arc-parametrized curve necessarily. In this example, $\alpha'(t) = (1, 3t^2, 0)$ reflecting the fact that $\alpha'(t) \neq 0$. It means that the curve has no singular points of order 0. It is a regular curve. One also sees that $\alpha''(t) = (0, 6t, 0)$ so that $\alpha''(0) = (0, 0, 0)$, i.e., 0 is a singular point of order 1.

In what follows, we restrict ourselves to arc-parametrized curve (which is automatically regular) with no singular point of order 1.

We shall denote by

$$t(s) = \alpha'(s) \tag{1.13}$$

the unit tangent vector of α at s so that

$$t'(s) = \alpha''(s) = \kappa(s) n(s) \tag{1.14}$$

The unit vector

$$t(s) \wedge {}^{3} n(s) = b(s). \tag{1.15}$$

is normal to the osculating plane and is called **binormal vector** at s.

$b'(s)$ measures the extent with which the curve is pulled out of the osculating plane. If $b'(s) = 0$, the curve always stays on the osculating plane and is never pulled out off it.

From $b(s) = t(s) \wedge n(s)$, one has

$$b'(s) = t'(s) \wedge n(s) + t(s) \wedge n'(s) = t(s) \wedge n'(s) \tag{1.16}$$

since $t'(s)$ and $n(s)$ are collinear.

In other words, $b'(s) \perp t(s)$. And $b'(s)$ is perpendicular to $b(s)$ itself. And the only way they are satisfied is $b'(s)$ is parallel to $n(s)$, i.e., $b'(s) \parallel n(s)$. And it prompts us to write

$$b'(s) = \tau(s) \cdot n(s) \tag{1.17}$$

for some function $\tau(s)$.

[3] Discussion on wedge/cross product of two vectors in $\mathbb{R}^3$ can be found in [**Car76**, p. 11].

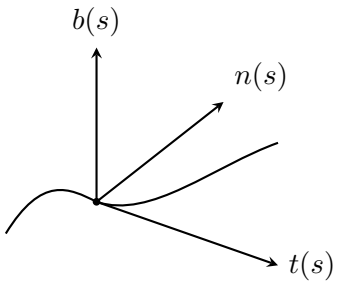


Figure 1.1: The Frenet trihedron at s.

Definition 1.2.4 (Torsion)**.** Let $\alpha : I \to \mathbb{R}^3$ be a curve parametrized by arc length s such that $\alpha''(s) \neq 0$, $s \in I$. The number $\tau(s)$ defined by $b'(s) = \tau(s)n(s)$ is called the **torsion of** α at s.

Remark 1.2.5. If α is a plane curve (i.e., $\alpha(I)$ is contained in a plane), then the plane of the curve agrees with the osculating plane; hence $\tau \equiv 0$. Conversely, if $\tau \equiv 0$ (and $\kappa \neq 0$) we have $b'(s) = 0$, so that

$$b(s) = b_0 = \text{constant}. \tag{1.18}$$

And therefore,

$$(\alpha(s) \cdot b_0)' = \alpha'(s) \cdot b_0 = 0 \tag{1.19}$$

since $b(s)$ is normal to $\alpha'(s) = t(s)$.

This implies

$$\alpha(s) \cdot b_0 = \text{constant}, \quad \forall s \in I. \tag{1.20}$$

We have $\alpha(s) \cdot b_0 = d =$ some constant for all $s \in I$. For an arbitrary reference point $s_0 \in I$, $[\alpha(s) - \alpha(s_0)] \cdot b_0 = 0$ meaning that the vector joining any two points on the curve $\alpha(s)$ will always be perpendicular to b_0. This means that the curve $\alpha(s)$ lies strictly in a plane perpendicular to b_0.

Note that $\kappa \neq 0$ is essential throughout as

$$n(s) = \frac{\alpha''(s)}{\kappa(s)} \tag{1.21}$$

if $\kappa(s) = 0$ anywhere $n(s)$ ceases to exist and since $b(s) = t(s) \wedge n(s)$, the existence of the constant vector b_0 collapses.

Summary. To each value of the parameter s, we have associated three orthogonal unit vectors $t(s), n(s), b(s)$. The trihedron thus formed is referred to as the **Frenet trihedron** at s. The derivatives $t'(s) = \kappa(s)n(s)$, $b'(s) = \tau(s)n(s)$ of the vectors $t(s)$ and $b(s)$, when expressed in the basis $\{t, n, b\}$, yield geometric entities curvature κ and torsion τ which give us information about the behaviour of α in the neighbourhood of s.

Let us now compute $n'(s)$ using $n(s) = b(s) \wedge t(s)$.

$$\begin{aligned} n'(s) &= b'(s) \wedge t(s) + b(s) \wedge t'(s) && (1.22)\\ &= \tau(s)(n(s) \wedge t(s)) + \kappa(s)(b(s) \wedge n(s)) && (1.23)\\ &= \tau(s)(-b(s)) + \kappa(s)(-t(s)) && (1.24)\\ &= -\tau(s)b(s) - \kappa(s)t(s). && (1.25) \end{aligned}$$

Here, we have used $b'(s) = \tau(s)n(s)$ and $t'(s) = \kappa(s)n(s)$.

We thus have **Frenet formulae**

$$\boxed{\begin{aligned} t'(s) &= \kappa(s)n(s) \\ n'(s) &= -\tau(s)b(s) - \kappa(s)t(s) \\ b'(s) &= \tau(s)n(s) \end{aligned}} \tag{1.26}$$

The tb plane is called **rectifying plane** and the nb plane is called the **normal plane**. The line that contains $n(s)$ and passes through $\alpha(s)$ is called the **principal normal**. And the line that contains $b(s)$ and passes through $\alpha(s)$ is called the **binormal**.

The inverse $R = \frac{1}{\kappa}$ of the curvature is called the **radius of curvature** at s.

Now, let us get back to the orientation discussion of the curve $\alpha(s)$. As $-s$ increases the original parameter s decreases. Define

$$\beta(-s) = \alpha(s). \tag{1.27}$$

so that

$$\beta'(-s) = \frac{d\beta}{d(-s)}(-s) = \frac{d\alpha(s)}{d(-s)} = -\frac{d\alpha(s)}{ds} = -\alpha'(s). \tag{1.28}$$

And

$$\beta''(-s) = \frac{d\beta'(-s)}{d(-s)} = \frac{d}{d(-s)}(-\alpha'(s)) \tag{1.29}$$

$$= \frac{d}{ds}(\alpha'(s)) = \alpha''(s). \tag{1.30}$$

Therefore,

$$\beta''(-s) = \alpha''(s). \tag{1.31}$$

Therefore, $\alpha'(s)$ changes sign if the orientation of the curve reverses while $\alpha''(s)$ and hence curvature remains invariant under orientation reversal of the curve.

Theorem 1.2.6 (Fundamental theorem of the local theory of curves)
Given differentiable functions $\kappa(s) > 0$ (recall $\kappa(s) = \|\alpha''(s)\|$ so that $\kappa(s) \geq 0$ and $\kappa(s) = 0$ is not allowed as $n(s)$ is defined through $n(s) = \frac{\alpha''(s)}{\kappa(s)}$) and $\tau(s)$, $s \in I$, there exists a regular parametrized curve $\alpha : I \to \mathbb{R}^3$ such that s is the arc length, $\kappa(s)$ is the curvature, and $\tau(s)$ is the torsion of α. (The regular parametrized curve here is actually arc length parametrized which is automatically regular—the existence of such an arc length parametrized curve is claimed in the fundamental theorem). Moreover, any other curve $\widetilde{\alpha}$ satisfying the same conditions, differs from α by a "rigid motion"; that is there exists an Orthogonal linear transformation ρ of $\mathbb{R}^3$, with positive determinant, and a vector c such that $\widetilde{\alpha} = \rho \circ \alpha + c$.

In the next class, we shall prove the fundamental theorem of local theory of curves.

2 Lecture 02

In this lecture we will prove Fundamental theorem of the local theory of curves. Before doing so, we will precisely define what we mean by an *orthogonal linear transformation* and prove some properties that will be needed in the proof.

§2.1 Orthogonal linear transformation

Notation. For $u, v \in \mathbb{R}^3$,

$$\langle u, v \rangle := u \cdot v.$$

Definition 2.1.1 (Orthogonal linear transformation)**.** A linear transformation $\rho : \mathbb{R}^3 \to \mathbb{R}^3$ is called *orthogonal* if

$$\langle \rho(u), \rho(v) \rangle = \langle u, v \rangle \tag{2.1}$$

for all $u, v \in \mathbb{R}^3$.

In particular, we have

$$\|\rho(u)\| = \|u\|. \tag{2.2}$$

Proposition 2.1.2 (Characterisation of orthogonal linear transformations)
Let A be the matrix representation of a linear transformation $\rho : \mathbb{R}^3 \to \mathbb{R}^3$. Then ρ is orthogonal if and only if

$$A^T A = A A^T = I_3. \tag{2.3}$$

Proof. ($\Leftarrow$) Suppose that $A^T A = A A^T = I_3$. Then, for all $u, v \in \mathbb{R}^3$,

$$\langle Au, Av \rangle = (Au)^T Av = u^T A^T A v = u^T I_3 v = \langle u, v \rangle \tag{2.4}$$

($\Rightarrow$) Conversely, suppose that ρ is orthogonal. Then, for all $u, v \in \mathbb{R}^3$,

$$\begin{aligned} &\langle Au, Av \rangle = \langle u, v \rangle \\ \Longrightarrow\ &u^T A^T A v = u^T v \\ \Longrightarrow\ &u^T (A^T A - I_3) v = 0, . \end{aligned} \tag{2.5}$$

for all $u, v \in \mathbb{R}^3$.

Taking $u = e_i$ and $v = e_j$, where $\{e_1, e_2, e_3\}$ is the standard basis of $\mathbb{R}^3$, we obtain

$$e_i^T (A^T A - I_3) e_j = 0$$

for every $i, j \in \{1, 2, 3\}$.

Hence every entry of $A^T A - I_3$ is zero, so

$$A^T A = I_3.$$

Now,

$$\det(A^T A) = \det(I_3) = 1 \implies \det(A^2) = 1 \implies \det(A) = \pm 1. \tag{2.6}$$

Thus A is invertible, it follows that $A^{-1} = A^T$, implying

$$AA^T = AA^{-1} = I_3$$

which completes the proof. □

Corollary 2.1.3

If $\rho : \mathbb{R}^3 \to \mathbb{R}^3$ is an orthogonal linear transformation, and A is the matrix representation of it, then

$$\det(A) = \pm 1 \tag{2.7}$$

Definition 2.1.4 (Rotation)**.** An orthogonal linear transformation of determinant $+1$ is often called a *rotation*.

Proposition 2.1.5

Let $\rho : \mathbb{R}^3 \to \mathbb{R}^3$ be a rotation. Then

$$\rho(u \wedge v) = \rho(u) \wedge \rho(v) \tag{2.8}$$

for all $u, v \in \mathbb{R}^3$.

Proof. Recall from [Car76, p. 12] that wedge product $u \wedge v$, where $u, v \in \mathbb{R}^3$, is characterised by the equation

$$(u \wedge v) \cdot w = \det(u, v, w) \qquad \forall w \in \mathbb{R}^3. \tag{2.9}$$

Let $u, v \in \mathbb{R}^3$. Then for any $w \in \mathbb{R}^3$

$$\begin{aligned} \langle \rho(u) \wedge \rho(v), \rho(w) \rangle &= \det(\rho(u), \rho(v), \rho(w)) \\ &= \det \rho \, \det(u, v, w) \\ &= (+1) \det(u, v, w) \\ &= \det(u, v, w). \end{aligned} \tag{2.10}$$

On the other hand,

$$\begin{aligned} \langle \rho(u \wedge v), \rho(w) \rangle &= \langle u \wedge v, w \rangle \qquad \text{(By 2.1)} \\ &= \det(u, v, w). \end{aligned} \tag{2.11}$$

Thus, the two vectors $\rho(u) \wedge \rho(v)$ and $\rho(u \wedge v)$ have the same inner product with every vector $\rho(w)$. Since ρ is surjective, they have the same inner product with every vector in $\mathbb{R}^3$, proving Equation 2.8. □

Definition 2.1.6 (Rigid motion)**.** Let $M : \mathbb{R}^3 \to \mathbb{R}^3$ be given by

$$M(x) = \rho(x) + c,$$

where ρ is an orthogonal linear transformation of determinant $+1$ (aka. a rotation) and $c \in \mathbb{R}^3$ is constant. Such a transformation is called an *orientation-preserving rigid motion*.

§2.2 Proof of the fundamental theorem

Proof of the uniqueness part of Fundamental theorem of the local theory of curves.

We shall first prove that the rigid motion $M : \mathbb{R}^3 \to \mathbb{R}^3$ preserves arc length, curvature, and torsion.

Arc length.

First take any regular parametrized curve $\gamma : J \to \mathbb{R}^3$ with parameter t, and put

$$\widehat{\gamma}(t) = M(\gamma(t)) = \rho(\gamma(t)) + c. \tag{2.12}$$

Since ρ is linear and c is constant,

$$\frac{d\widehat{\gamma}}{dt} = \rho\left(\frac{d\gamma}{dt}\right).$$

Then by Equation 2.2, one has

$$\left\|\frac{d\widehat{\gamma}}{dt}\right\| = \left\|\frac{d\gamma}{dt}\right\|. \tag{2.13}$$

Integrating over any $[a, b] \subset J$, one has

$$\int_a^b \left\|\frac{d\widehat{\gamma}}{dt}\right\| dt = \int_a^b \left\|\frac{d\gamma}{dt}\right\| dt. \tag{2.14}$$

Moreover, if we choose the same base parameter t_0, the arc-length functions agree:

$$\widehat{s}(t) = \int_{t_0}^t \left\|\frac{d\widehat{\gamma}}{du}\right\| du = \int_{t_0}^t \left\|\frac{d\gamma}{du}\right\| du = s(t). \tag{2.15}$$

Now, since the curves are regular,

$$s'(t) = \|\gamma'(t)\| > 0, \tag{2.16}$$

so that $s(t)$ has inverse $t = t(s)$, where, by an abuse of notation, t also denotes the inverse function s^{-1} of s. Because $\widehat{s}(t)$ is the same function as $s(t)$, their inverses also agree:

$$\widehat{t}(s) = t(s). \tag{2.17}$$

Now, define the arc-length parametrizations by

$$\Gamma(s) = \gamma \circ t(s) = \gamma(t(s)), \qquad \widehat{\Gamma}(s) = \widehat{\gamma}(\widehat{t}(s)). \tag{2.18}$$

Then,

$$\begin{aligned}
\widehat{\Gamma}(s) &= \widehat{\gamma}(\widehat{t}(s)) \\
&= \widehat{\gamma}(t(s)); \text{[By 2.17]} \\
&= M(\gamma(t(s))); \text{[By 2.12]} \\
&= M(\Gamma(s)); \text{[By 2.18]}.
\end{aligned}$$

If we abuse notation and reuse the symbols $\gamma, \widehat{\gamma}$ for their arc-length parametrizations instead of Γ and $\widehat{\Gamma}$, we have

$$\widehat{\gamma}(s) = M(\gamma(s)). \tag{2.19}$$

Now, we check curvature and torsion invariance using unit-speed/arc-length parametrizations.

Curvature.

Taking the derivative of Equation 2.19,

$$\widehat{\gamma}'(s) = \rho(\gamma'(s)). \tag{2.20}$$

$\gamma(s)$ is of unit speed, i.e., $|\gamma'(s)| = 1$. In other words, $\gamma'(s)$ is a vector of unit length so that the rotated vector $\rho(\gamma'(s)) = \widehat{\gamma}'(s)$ will also be of unit length, i.e., $\widehat{\gamma}(s)$ is also of unit speed.

Differentiating Equation 2.20 once more,

$$\widehat{\gamma}''(s) = \rho(\gamma''(s)). \tag{2.21}$$

Therefore,

$$\begin{aligned} k_{\widehat{\gamma}}(s) &= \|\widehat{\gamma}''(s)\| \\ &= \|\rho(\gamma''(s))\|; \text{[By 2.21]} \\ &= \|\gamma''(s)\|; \text{[By 2.2]} \\ &= k_\gamma(s). \end{aligned} \tag{2.22}$$

The Frenet frame.

Assume $k_\gamma > 0$. The tangent vectors satisfy (we're using $\{T, N, B\}$ $\{\widehat{T}, \widehat{N}, \widehat{B}\}$ as Frenet frame for the curves γ and $\widehat{\gamma}$, respectively).

$$\begin{aligned} \widehat{T}(s) = \widehat{\gamma}'(s) &= \rho(\gamma'(s)); \text{[By 2.20]} \\ &= \rho(T(s)). \end{aligned} \tag{2.23}$$

Using curvature invariance Equation 2.22,

$$\begin{aligned} \widehat{N}(s) = \frac{\widehat{\gamma}''(s)}{k_{\widehat{\gamma}}(s)} &= \frac{\rho(\gamma''(s))}{k_\gamma(s)} && \text{[Using (2.21) and (2.22)]} \\ &= \rho\left(\frac{\gamma''(s)}{k_\gamma(s)}\right) = \rho(N(s)). \end{aligned} \tag{2.24}$$

Finally,

$$\begin{aligned} \widehat{B}(s) &= \widehat{T}(s) \wedge \widehat{N}(s) \\ &= \rho(T(s)) \wedge \rho(N(s)) && \text{[By 2.23 and 2.24]} \\ &= \rho(T(s) \wedge N(s)) && \text{[By 2.2]} \\ &= \rho(B(s)). \end{aligned} \tag{2.25}$$

Thus, we have

$$\boxed{\begin{aligned} \widehat{T}(s) &= \rho(T(s)), \\ \widehat{N}(s) &= \rho(N(s)), \\ \widehat{B}(s) &= \rho(B(s)). \end{aligned}} \tag{2.26}$$

Torsion.

Differentiate the last of the boxed set of equations given by Equation 2.26:

$$\widehat{B}'(s) = \rho(B'(s)). \tag{2.27}$$

Since $B'(s) = \tau_\gamma(s)N(s)$, one has

$$\widehat{B}'(s) = \rho(\tau_\gamma(s)N(s))$$

$$
\begin{aligned}
&= \tau_\gamma(s)\rho(N(s)) \\
&= \tau_\gamma(s)\widehat{N}(s), \qquad \text{[By the second of Equation 2.26]}
\end{aligned} \tag{2.28}
$$

But torsion of $\widehat{\gamma}(s)$ is defined by

$$\widehat{B}'(s) = \tau_{\widehat{\gamma}}(s)\widehat{N}(s). \tag{2.29}$$

Comparing Equation 2.28 and Equation 2.29, one gets

$$\tau_\gamma(s) = \tau_{\widehat{\gamma}}(s). \tag{2.30}$$

Align the initial points and Frenet frames.

Now, let $\alpha, \bar{\alpha} : I \to \mathbb{R}^3$ be two arc-parametrized curves having the same curvature and torsion:

$$k_\alpha(s) = k_{\bar{\alpha}}(s) = k(s) > 0, \qquad \tau_\alpha(s) = \tau_{\bar{\alpha}}(s) = \tau(s). \tag{2.31}$$

Fix $s_0 \in I$. Write their initial Frenet frames as $\{T_0, N_0, B_0\}$ and $\{\bar{T}_0, \bar{N}_0, \bar{B}_0\}$. We first justify that a rotation can carry the second frame to the first. Form the matrices whose columns are these vectors:

$$F_0 = [T_0 \ N_0 \ B_0], \qquad \bar{F}_0 = [\bar{T}_0 \ \bar{N}_0 \ \bar{B}_0]. \tag{2.32}$$

at the fixed parameter value s_0,

$$T_0 = T(s_0) = \begin{bmatrix} T_1 \\ T_2 \\ T_3 \end{bmatrix}, \quad N_0 = N(s_0) = \begin{bmatrix} N_1 \\ N_2 \\ N_3 \end{bmatrix}, \quad B_0 = B(s_0) = \begin{bmatrix} B_1 \\ B_2 \\ B_3 \end{bmatrix}. \tag{2.33}$$

An interlude (On orientation of an ordered basis).

Two ordered bases (a, b, c) and (a', b', c') of $\mathbb{R}^3$ have the same orientations if the change of basis matrix from (a, b, c) to (a', b', c') has positive determinant.

We have $F_0 = [T_0 \ N_0 \ B_0]$ and the standard coordinate frame $\mathcal{E} = [e_1, e_2, e_3]$.

Express the Frenet vectors in the standard basis.

$$T_0 = \sum_{i=1}^{3} T_i e_i, \qquad N_0 = \sum_{i=1}^{3} N_i e_i, \qquad B_0 = \sum_{i=1}^{3} B_i e_i. \tag{2.34}$$

Then

$$F_0 = \begin{bmatrix} T_1 & N_1 & B_1 \\ T_2 & N_2 & B_2 \\ T_3 & N_3 & B_3 \end{bmatrix}, \tag{2.35}$$

so that the columns of F_0 are the standard coordinates of the Frenet basis vectors.

Now, take $v \in \mathbb{R}^3$, and suppose

$$v = x_1 T_0 + x_2 N_0 + x_3 B_0. \tag{2.36}$$

Its Frenet coordinates are

$$[v]_{F_0} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}. \tag{2.37}$$

Then

$$v = x_1(T_1 e_1 + T_2 e_2 + T_3 e_3) + x_2(N_1 e_1 + N_2 e_2 + N_3 e_3)$$

$$
\begin{aligned}
&+ x_3(B_1e_1 + B_2e_2 + B_3e_3) \qquad \text{[substituting (2.34) into (2.36)]} \\
&= (x_1T_1 + x_2N_1 + x_3B_1)e_1 + (x_1T_2 + x_2N_2 + x_3B_2)e_2 \\
&\quad + (x_1T_3 + x_2N_3 + x_3B_3)e_3.
\end{aligned}
$$

So that

$$
\begin{aligned}
[v]_{\mathcal{E}} &= \begin{bmatrix} x_1T_1 + x_2N_1 + x_3B_1 \\ x_1T_2 + x_2N_2 + x_3B_2 \\ x_1T_3 + x_2N_3 + x_3B_3 \end{bmatrix} \\
&= \begin{bmatrix} T_1 & N_1 & B_1 \\ T_2 & N_2 & B_2 \\ T_3 & N_3 & B_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\
&= F_0[v]_{F_0}.
\end{aligned}
$$

Therefore,

$$
[v]_{\mathcal{E}} = F_0[v]_{F_0}. \tag{2.38}
$$

In other words, F_0 takes coordinates relative to F_0 to coordinates relative to $\mathcal{E}$.

The Frenet frame is orthonormal.

$\|T_0\| = \|N_0\| = \|B_0\| = 1$ and $\langle T_0, N_0\rangle = \langle T_0, B_0\rangle = \langle N_0, B_0\rangle = 0$.

$$
\begin{aligned}
\therefore \quad F_0^T F_0 &= \begin{bmatrix} T_1 & T_2 & T_3 \\ N_1 & N_2 & N_3 \\ B_1 & B_2 & B_3 \end{bmatrix} \begin{bmatrix} T_1 & N_1 & B_1 \\ T_2 & N_2 & B_2 \\ T_3 & N_3 & B_3 \end{bmatrix} \\
&= \begin{bmatrix} \|T_0\|^2 & \langle T_0, N_0\rangle & \langle T_0, B_0\rangle \\ \langle N_0, T_0\rangle & \|N_0\|^2 & \langle N_0, B_0\rangle \\ \langle B_0, T_0\rangle & \langle B_0, N_0\rangle & \|B_0\|^2 \end{bmatrix} \\
&= I_3.
\end{aligned} \tag{2.39}
$$

Now, calculate $\det F_0 = \det(T_0, N_0, B_0) = \langle T_0 \wedge N_0, B_0\rangle = \langle B_0, B_0\rangle = 1 > 0$.

Therefore, F_0 has the same orientation as that of $\mathcal{E}$.

Similarly, for $\bar{F}_0$, $\bar{F}_0\bar{F}_0^T = \bar{F}_0^T\bar{F}_0 = I_3$, with $\det \bar{F}_0 = 1$.

Define ρ to have matrix form

$$
R = F_0\bar{F}_0^T. \tag{2.40}
$$

Then

$$
R^TR = \bar{F}_0F_0^TF_0\bar{F}_0^T = \bar{F}_0\bar{F}_0^T = I_3,
$$

and $\det R = \det F_0 \det \bar{F}_0^T = 1$. Thus ρ is a rotation. Moreover, from Equation 2.40, one has

$$
\begin{aligned}
R(\bar{F}_0^T)^{-1} = F_0 &\implies R(\bar{F}_0^{-1})^T = F_0 \\
&\implies R(\bar{F}_0^T)^T = F_0 \implies R\bar{F}_0 = F_0.
\end{aligned} \tag{2.41}
$$

which says precisely that

$$
\boxed{\begin{aligned} \rho(\bar{T}_0) &= T_0, \\ \rho(\bar{N}_0) &= N_0, \\ \rho(\bar{B}_0) &= B_0. \end{aligned}} \tag{2.42}
$$

The points p and $\bar{p}$ are the initial points. In our case, for the fixed rotation ρ, $M : \mathbb{R}^3 \to \mathbb{R}^3$ is the unique orientation-preserving rigid motion with rotational part ρ satisfying $M(\bar{p}) = p$ (see Remark 2.2.1), which translates to

$$M(\bar{\alpha}(s_0)) = \alpha(s_0). \tag{2.43}$$

Now, introduce

$$\beta(s) = M(\bar{\alpha}(s)). \tag{2.44}$$

By the invariance results proved earlier, the arc parametrized curve $\beta(s)$ has curvature $k(s)$ and torsion $\tau(s)$. Denote its Frenet frame by $\{\widetilde{T}(s), \widetilde{N}(s), \widetilde{B}(s)\}$. Now, rewrite Equation 2.44 as

$$\beta(s) = \rho(\bar{\alpha}(s)) + c, \qquad \text{where } c \text{ is a constant.}$$

Then

$$\beta'(s) = \rho(\bar{\alpha}'(s)). \tag{2.45}$$

$$\implies \widetilde{T}(s) = \rho(\bar{T}(s)). \tag{2.46}$$

But we already know that (By Equation 2.42)

$$\rho(\bar{T}(s_0)) = T(s_0). \tag{2.47}$$

And taking $s = s_0$ in Equation 2.46, one has

$$\widetilde{T}(s_0) = \rho(\bar{T}(s_0)). \tag{2.48}$$

Combining the previous two equations, one obtains

$$\widetilde{T}(s_0) = T(s_0). \tag{2.49}$$

Now, differentiating Equation 2.45 once again,

$$\beta''(s) = \rho(\bar{\alpha}''(s)). \tag{2.50}$$

Orthogonality of ρ yields

$$\|\beta''(s)\| = \|\rho(\bar{\alpha}''(s))\| = \|\bar{\alpha}''(s)\| = k(s) > 0. \tag{2.51}$$

Therefore,

$$\begin{aligned}
\widetilde{N}(s_0) &= \frac{\beta''(s_0)}{\|\beta''(s_0)\|} \\
&= \frac{\rho(\bar{\alpha}''(s_0))}{k(s_0)} \qquad \text{[using Equation 2.50 and Equation 2.51]} \\
&= \rho\left(\frac{\bar{\alpha}''(s_0)}{k(s_0)}\right) \\
&= \rho(\bar{N}(s_0)) = N(s_0) \qquad \text{[By Equation 2.42].}
\end{aligned}$$

$$\widetilde{N}(s_0) = N(s_0). \tag{2.52}$$

Now, using Equation 2.49 and Equation 2.52, one obtains the following:

$$\widetilde{B}(s_0) = \widetilde{T}(s_0) \wedge \widetilde{N}(s_0) = T(s_0) \wedge N(s_0) = B(s_0). \tag{2.53}$$

Recall our rigid motion expression $M(x) = \rho(x) + c$ with $c = p - \rho(\bar{p})$, where $p = \alpha(s_0)$ and $\bar{p} = \bar{\alpha}(s_0)$, so that

$$c = \alpha(s_0) - \rho(\bar{\alpha}(s_0)). \tag{2.54}$$

Consequently,

$$\begin{aligned}\beta(s_0) &= \rho(\bar{\alpha}(s_0)) + c \\ &= \rho(\bar{\alpha}(s_0)) + \alpha(s_0) - \rho(\bar{\alpha}(s_0)) \qquad \text{[using Equation 2.54]} \\ &= \alpha(s_0).\end{aligned}$$

$$\beta(s_0) = \alpha(s_0). \tag{2.55}$$

Prove that the aligned frames agree everywhere.

The Frenet frames of α and β satisfy the Frenet equations with the same coefficients k, τ:

$$\begin{aligned} T' &= kN, & \widetilde{T}' &= k\widetilde{N}, \\ N' &= -kT - \tau B, & \widetilde{N}' &= -k\widetilde{T} - \tau\widetilde{B}, \\ B' &= \tau N, & \widetilde{B}' &= \tau\widetilde{N}. \end{aligned} \tag{2.56}$$

Define the vector differences

$$\mathcal{U} = T - \widetilde{T}, \qquad \mathcal{V} = N - \widetilde{N}, \qquad \mathcal{W} = B - \widetilde{B}. \tag{2.57}$$

Then, subtracting the Frenet equations in Equation 2.56 yields

$$\begin{aligned} (T - \widetilde{T})' &= T' - \widetilde{T}' = k(N - \widetilde{N}), \\ (N - \widetilde{N})' &= N' - \widetilde{N}' = -k(T - \widetilde{T}) - \tau(B - \widetilde{B}), \\ (B - \widetilde{B})' &= B' - \widetilde{B}' = \tau(N - \widetilde{N}). \end{aligned} \tag{2.58}$$

Now, the above set of equations can be cast into the following form using $\mathcal{U}, \mathcal{V}, \mathcal{W}$ and their derivatives:

$$\mathcal{U}' = k\mathcal{V}, \qquad \mathcal{V}' = -k\mathcal{U} - \tau\mathcal{W}, \qquad \mathcal{W}' = \tau\mathcal{V}. \tag{2.59}$$

Now, consider the nonnegative function

$$E(s) = \frac{1}{2}\left(\|\mathcal{U}(s)\|^2 + \|\mathcal{V}(s)\|^2 + \|\mathcal{W}(s)\|^2\right).$$

Now, for the differentiable vector valued function $\mathcal{U}(s)$,

$$\frac{d}{ds}\|\mathcal{U}(s)\|^2 = \frac{d}{ds}\langle \mathcal{U}, \mathcal{U}\rangle = 2\langle \mathcal{U}, \mathcal{U}'\rangle. \tag{2.60}$$

Applying this identity to all 3 terms and adding them,

$$\frac{d}{ds}\left(\frac{1}{2}\left(\|\mathcal{U}(s)\|^2 + \|\mathcal{V}(s)\|^2 + \|\mathcal{W}(s)\|^2\right)\right) = \langle \mathcal{U}, \mathcal{U}'\rangle + \langle \mathcal{V}, \mathcal{V}'\rangle + \langle \mathcal{W}, \mathcal{W}'\rangle. \tag{2.61}$$

Substituting Equation 2.59 in Equation 2.61, one obtains,

$$\begin{aligned} E'(s) &= \langle \mathcal{U}, k\mathcal{V}\rangle + \langle \mathcal{V}, -k\mathcal{U} - \tau\mathcal{W}\rangle + \langle \mathcal{W}, \tau\mathcal{V}\rangle \\ &= k\langle \mathcal{U}, \mathcal{V}\rangle - k\langle \mathcal{V}, \mathcal{U}\rangle - \tau\langle \mathcal{V}, \mathcal{W}\rangle + \tau\langle \mathcal{W}, \mathcal{V}\rangle \\ &= 0. \end{aligned}$$

Hence, $E(s)$ is constant on the interval I.

Now, using Equation 2.57, one has

$$\begin{aligned} \mathcal{U}(s_0) &= T(s_0) - \widetilde{T}(s_0) = 0 \quad \text{[using Equation 2.49]}, \\ \text{Similarly,} \quad \mathcal{V}(s_0) &= \mathcal{W}(s_0) = 0. \end{aligned} \tag{2.62}$$

From Equation 2.62, one has,

$$E(s_0) = 0. \tag{2.63}$$

But $E'(s) = 0 \implies E(s) = 0$ for every $s \in I$.

$$E(s) = 0 \quad \text{for every } s \in I. \tag{2.64}$$

But each summand in

$$E(s) = \frac{1}{2}\left(\|\mathcal{U}(s)\|^2 + \|\mathcal{V}(s)\|^2 + \|\mathcal{W}(s)\|^2\right)$$

is nonnegative. Their sum vanish implies that each summand vanishes; thus,

$$\begin{aligned} \mathcal{U}(s) = T(s) - \widetilde{T}(s) = 0 &\implies T(s) = \widetilde{T}(s), \\ \text{Similarly,} \quad N(s) = \widetilde{N}(s), &\quad B(s) = \widetilde{B}(s), \quad \forall s \in I. \end{aligned} \tag{2.65}$$

Prove that the aligned curves agree, then undo the rigid motion.

We know that $\alpha'(s) = T(s)$ and $\beta'(s) = \widetilde{T}(s)$. Then by Equation 2.65, $(\alpha - \beta)' = 0$. Therefore, $\alpha - \beta$ is constant. Using Equation 2.55, one obtains

$$\alpha(s) - \beta(s) = \alpha(s_0) - \beta(s_0) = 0.$$

Hence,

$$\alpha(s) = \beta(s), \qquad \forall s \in I. \tag{2.66}$$

But recall Equation 2.44 that $\beta = M \circ \bar{\alpha}$.

$$\beta = M \circ \bar{\alpha} \implies \alpha = M \circ \bar{\alpha} \qquad \text{[using Equation 2.66]}. \tag{2.67}$$

Consequently,

$$\bar{\alpha} = M^{-1} \circ \alpha. \tag{2.68}$$

We've seen $\alpha(s) = \beta(s) = \rho(\bar{\alpha}(s)) + c$. Then

$$\alpha(s) - c = \rho(\bar{\alpha}(s)). \tag{2.69}$$

Since ρ is a rotation, it is invertible. Apply ρ^{-1} on both sides of Equation 2.69.

$$\rho^{-1}(\alpha(s) - c) = \rho^{-1}(\rho(\bar{\alpha}(s))) = \bar{\alpha}(s). \tag{2.70}$$

Finally, ρ^{-1} is linear so that

$$\rho^{-1}(\alpha(s) - c) = \rho^{-1}(\alpha(s)) - \rho^{-1}(c). \tag{2.71}$$

Comparing Equation 2.70 and Equation 2.71, one obtains,

$$\bar{\alpha}(s) = \rho^{-1}(\alpha(s)) - \rho^{-1}(c). \tag{2.72}$$

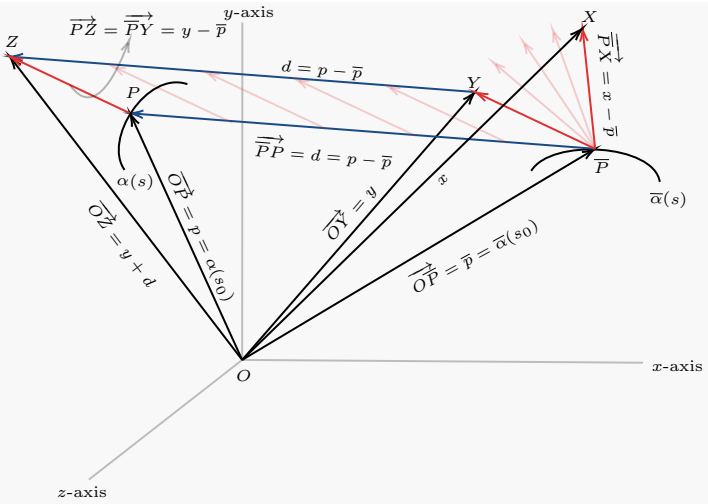
Equation 2.68 reveals the fact that the two curves α and $\bar{\alpha}$ are related by a rigid motion. □

Remark 2.2.1. Let $\bar{p}$ and p be the initial points of the two curves $\alpha, \overline{\alpha}$, and fix a rotation ρ. We wish to construct the unique orientation-preserving rigid motion $M : \mathbb{R}^3 \to \mathbb{R}^3$ with rotational part ρ such that $M(\bar{p}) = p$.

Rotating in $\mathbb{R}^3$ about $\bar{p} \in \mathbb{R}^3$ is given by

$$R_{\bar{p}}(x) = \bar{p} + \rho(x - \bar{p})^{a}. \tag{2.73}$$

To understand this more explicitly, consider the following picture:



Rotation of x about $\bar{p}$, followed by translation by $d = p - \bar{p}$.

1. Let $\bar{P}$ denote the tip of the vector $\bar{p}$ starting at O. The corresponding geometric vector is denoted by $\overrightarrow{OP} = \bar{p}$. Similarly, let X denote the tip of the vector x starting at O, so that $\overrightarrow{OX} = x$. $\overrightarrow{PX} = x - \bar{p}$ under rotation about $\bar{P}$, $\overrightarrow{PX}$ is rotated to $\overrightarrow{PY} = \rho(x - \bar{p})$ whose tip is at Y. Hence,

$$\begin{aligned} y = \overrightarrow{OY} &= \overrightarrow{OP} + \overrightarrow{PY} \\ &= \bar{p} + \rho(x - \bar{p}). \end{aligned} \tag{2.74}$$

2. Translate by $d = p - \bar{p}$, which is represented by both $\overrightarrow{PP}$ and its parallel translate $\overrightarrow{YZ}$. We have

$$\begin{aligned} \overrightarrow{PZ} &= \overrightarrow{OZ} - \overrightarrow{OP} \\ &= (y + d) - (\bar{p} + d) \\ &= y - \bar{p}. \end{aligned}$$

   Also, from the triangle $O\bar{P}Y$, one has

$$\overrightarrow{PY} = y - \bar{p} = \overrightarrow{PZ}.$$

   Finally, Z is the image of X under the full rigid motion. Its position vector from O is

$$\begin{aligned} \overrightarrow{OZ} &= y + d \\ &= \bar{p} + \rho(x - \bar{p}) + p - \bar{p} \\ &= p + \rho(x - \bar{p}) \\ &= M(x), \end{aligned} \tag{2.75}$$

   where we have used (2.74) and the fact that $d = p - \bar{p}$.

Thus, the rigid motion is given by

$$M(x) = p + \rho(x - \bar{p}) = \rho(x) + c, \qquad c = p - \rho(\bar{p}). \tag{2.76}$$

In particular, plugging in $x = \bar{p}$ gives $M(\bar{p}) = p$.

Moreover, every orientation-preserving rigid motion with rotational part ρ has the form $\rho(x) + c$. Since the condition $M(\bar{p}) = p$ uniquely determines $c = p - \rho(\bar{p})$, the constructed motion is the unique rigid motion with the prescribed rotational part ρ that maps $\bar{p}$ to p.

[a] Aside: $R_{\bar{p}}(\bar{p}) = \bar{p}$ as rotation about $\bar{p}$ keeps the vector $\bar{p}$ fixed. But $\rho(\bar{p})$ is the rotated vector after one rotates the vector $\bar{p}$ about the origin and hence $\rho(\bar{p}) \neq \bar{p}$. But indeed, $\rho(0) = 0$.

We will prove the existence part of the Fundamental Theorem in the next class.

3 Lecture 03

§3.1 Proof of the fundamental theorem (continued)

Proof of existence part of Fundamental theorem of the local theory of curves.

Choose s_0, a point $p_0 \in \mathbb{R}^3$, and a positively orientated orthonormal initial frame $\{T_0, N_0, B_0\}$ with $F_0 = \begin{bmatrix} T_0 & N_0 & B_0 \end{bmatrix}$ being an orthonormal matrix of unit determinant. We now have to construct a curve from prescribed $\kappa(s) > 0$ and $\tau(s)$.

Define

$$A(s) = \begin{bmatrix} 0 & -\kappa(s) & 0 \\ \kappa(s) & 0 & \tau(s) \\ 0 & -\tau(s) & 0 \end{bmatrix}, \tag{3.1}$$

where $A(s)^T = -A(s)$.

We have the matrix differential equation

$$F'(s) = F(s)A(s) \qquad \text{with } F(s_0) = F_0. \tag{3.2}$$

Explicitly,

$$\begin{bmatrix} T_1'(s) & N_1'(s) & B_1'(s) \\ T_2'(s) & N_2'(s) & B_2'(s) \\ T_3'(s) & N_3'(s) & B_3'(s) \end{bmatrix} = \begin{bmatrix} T_1(s) & N_1(s) & B_1(s) \\ T_2(s) & N_2(s) & B_2(s) \\ T_3(s) & N_3(s) & B_3(s) \end{bmatrix} \begin{bmatrix} 0 & -\kappa(s) & 0 \\ \kappa(s) & 0 & \tau(s) \\ 0 & -\tau(s) & 0 \end{bmatrix} \tag{3.3}$$

$$\implies \begin{bmatrix} T_1'(s) & N_1'(s) & B_1'(s) \\ T_2'(s) & N_2'(s) & B_2'(s) \\ T_3'(s) & N_3'(s) & B_3'(s) \end{bmatrix} = \begin{bmatrix} \kappa(s)N_1(s) & -\kappa(s)T_1(s) - \tau(s)B_1(s) & \tau(s)N_1(s) \\ \kappa(s)N_2(s) & -\kappa(s)T_2(s) - \tau(s)B_2(s) & \tau(s)N_2(s) \\ \kappa(s)N_3(s) & -\kappa(s)T_3(s) - \tau(s)B_3(s) & \tau(s)N_3(s) \end{bmatrix}.$$

By equating the columns, one has

$$\begin{aligned} T'(s) &= \kappa(s)N(s) \\ N'(s) &= -\kappa(s)T(s) - \tau(s)B(s) \\ B'(s) &= \tau(s)N(s). \end{aligned} \tag{3.4}$$

In other words, we are looking for the unique solution of the linear differential system with continuous coefficients with the initial value

$$F(s_0) = \begin{bmatrix} T(s_0) & N(s_0) & B(s_0) \end{bmatrix} = \begin{bmatrix} T_0 & N_0 & B_0 \end{bmatrix} = F_0.$$

Now

$$\begin{aligned} \frac{d}{ds}\left(F(s)F(s)^T\right) &= F'(s)F(s)^T + F(s)(F'(s))^T \\ &= F(s)A(s)F(s)^T + F(s)A^T(s)F(s)^T \qquad \text{[By (3.2)]} \\ &= F(s)\left(A(s) + A^T(s)\right)F(s)^T \\ &= 0; \qquad [\because A^T(s) = -A(s)] \end{aligned}$$

Thus

$$F(s)F(s)^T = F(s_0)F(s_0)^T = F_0F_0^T = I_3. \tag{3.5}$$

Therefore, $F^{-1}(s) = F(s)^T$. Hence $F(s)F(s)^T = F(s)^TF(s) = I_3$. Since $F(s)$ is an orthogonal matrix, its columns are orthonormal.

Now the function

$$g(s) = \det F(s). \tag{3.6}$$

is continuous because the entries of $F(s)$ are continuous and the determinant is a polynomial in these entries. We know $g(s) \in \{-1, 1\}$ for every $s \in I$. And $g(s_0) = 1$ as we initially chose.

If $g(s) = -1$ anywhere in I, the intermediate value theorem on the interval between s_0 and s_1 would give some s_* with $g(s_*) = 0$, which is a contradiction! Hence,

$$\det F(s) = 1 \qquad \forall s \in I^{1}. \tag{3.7}$$

Now define

$$\alpha(s) = p_0 + \int_{s_0}^{s} T(u)\, du. \tag{3.8}$$

Then by the fundamental theorem of calculus,

$$\alpha'(s) = T(s). \tag{3.9}$$

And since $\{T(s), N(s), B(s)\}$ constitutes an orthonormal basis at each parameter value $s \in I$,

$$\|\alpha'(s)\| = \|T(s)\| = 1. \tag{3.10}$$

Thus the curve $\alpha(s)$ defined by (3.8) is arc-length parametrized, and hence regular.

Furthermore,

$$\alpha''(s) = T'(s) = \kappa(s)N(s); \qquad \text{[By (3.4)]} \tag{3.11}$$

Since $\|N(s)\| = 1$ by orthonormality of the basis $\{T(s), N(s), B(s)\}$, and $\kappa(s) > 0$ for all $s \in I$, one obtains

$$\|\alpha''(s)\| = \kappa(s) \qquad \text{and} \qquad \frac{\alpha''(s)}{\|\alpha''(s)\|} = N(s). \tag{3.12}$$

Hence, the curvature of $\alpha(s)$ is indeed prescribed by $\kappa(s)$ and its principal normal is the constructed vector $N(s)$. Now, since $\{T(s), N(s), B(s)\}$ constitutes an orthonormal system and $\det F(s) = 1$, one must have

$$B(s) = T(s) \wedge N(s). \tag{3.13}$$

So, the binormal of the curve $\alpha(s)$ is the constructed vector $B(s)$.

Finally, by (3.4), we have $B'(s) = \tau(s)N(s)$, so the torsion of $\alpha(s)$ is precisely the prescribed function $\tau(s)$.

This proves existence and completes the fundamental theorem. □

[1] Note that $s_0 \in I$ is just a prescribed parameter value at which the initial data $F(s_0)$ is given. It doesn't mean that the parameter s is supposed to be always greater or equal to s_0.

4 Lecture 04

Notes of this lecture is still somewhat rough, as I haven't had the chance to proofread everything thoroughly, and a few pictures still need to be added. Other than that, I've copied almost everything directly from the handwritten notes and given it a quick review.

§4.1 The local canonical form

The key idea is to fix the Frenet tripod at one point and use it as a coordinate system for the nearby curve.

Let $\alpha : I \to \mathbb{R}^3$ be a curve parametrised by arc length without singular points of order 1. k is a differentiable function with $k(s_0) > 0$. By continuity of k, we have $k > 0$ in a neighborhood of s_0, so that the Frenet frame is defined there. Put $h = s - s_0$ and write

$$\widetilde{\alpha}(h) = \alpha(s_0 + h). \tag{4.1}$$

Then

$$\widetilde{\alpha}'(h) = \alpha'(s_0 + h)\frac{d}{dh}(s_0 + h) = \alpha'(s_0 + h). \tag{4.2}$$

Thus $\|\widetilde{\alpha}'(h)\| = 1$, since $\|\alpha'(s)\| = 1$ as α is arc-parametrized.

Write $p_0 = \alpha(0)$ (we dropped the tilde and we'll be denoting s by h from now on), and

$$T_0 = T(0), \qquad N_0 = N(0), \qquad B_0 = B(0). \tag{4.3}$$

We place the spatial origin at p_0 and use T_0, N_0, B_0 as the positive x, y, z directions, respectively.

Diagram: To be drawn.

The source tripod $\{T_0, N_0, B_0\}$ is located at $p_0 = \alpha(0)$, while the destination tripod $\{e_1, e_2, e_3\}$ is located at the origin O, with $\beta(0) = O$. In the uniqueness proof, the source tripod $\{\overline{T}_0, \overline{N}_0, \overline{B}_0\}$ was transported to the destination tripod $\{T_0, N_0, B_0\}$ by a rotation ρ represented by the matrix R, with $R = F_0\overline{F}_0^{\mathsf{T}}$, where $F_0 = [T_0, N_0, B_0]$ and $\overline{F}_0 = [\overline{T}_0, \overline{N}_0, \overline{B}_0]$ are 3×3 orthogonal matrices with unit determinant. Here, $\rho(\overline{T}_0) = T_0$, $\rho(\overline{N}_0) = N_0$, $\rho(\overline{B}_0) = B_0$, with $M(\overline{\alpha}(s)) = \alpha(s)$.

In the present case, $M(p_0) = O$, and

$$\rho(T_0) = e_1, \qquad \rho(N_0) = e_2, \qquad \rho(B_0) = e_3. \tag{4.4}$$

And the rotation matrix $R = F_0^{\mathsf{T}}$, simply with $F_0 = [T_0, N_0, B_0]$ (unbarred). For our present purpose, we denote the matrix F_0 by S, so that

$$R = S^{\mathsf{T}}, \qquad S = [T_0, N_0, B_0]. \tag{4.5}$$

We'll call $\beta(s)$, with $\beta(0) = O$, the "normalized curve" defined by

$$\beta(s) = M(\alpha(s)) = \begin{bmatrix} x(s) \\ y(s) \\ z(s) \end{bmatrix}. \tag{4.6}$$

We first rotate about $p_0 = \alpha(0)$ and then translate it to the origin:

$$R_{p_0}(x) = p_0 + \rho(x - p_0). \tag{4.7}$$

Thus $\xi_1(s) = p_0 + \rho(\alpha(s) - p_0)$ (rotate about p_0), and

$$\begin{aligned} \beta(s) &= \xi_1(s) - p_0 \qquad \text{(translate by } -p_0\text{)} \\ &= p_0 + \rho(\alpha(s) - p_0) - p_0 = \rho(\alpha(s) - p_0). \end{aligned} \tag{4.8}$$

Now, $x(s) = \langle \beta(s), e_1 \rangle = \langle \rho(\alpha(s) - p_0), \rho(T_0) \rangle = \langle \alpha(s) - p_0, T_0 \rangle$, because ρ is orthogonal. Similarly, $y(s) = \langle \alpha(s) - p_0, N_0 \rangle$ and $z(s) = \langle \alpha(s) - p_0, B_0 \rangle$. Together,

$$\begin{bmatrix} x(s) \\ y(s) \\ z(s) \end{bmatrix} = \begin{bmatrix} \langle \alpha(s) - p_0, T_0 \rangle \\ \langle \alpha(s) - p_0, N_0 \rangle \\ \langle \alpha(s) - p_0, B_0 \rangle \end{bmatrix}. \tag{4.9}$$

We have

$$\alpha'(s) = T(s), \qquad \alpha''(s) = T'(s) = k(s)N(s), \tag{4.10}$$

and

$$N'(s) = -\tau(s)B(s) - k(s)T(s). \tag{4.11}$$

Now, differentiate $\alpha''(s) = k(s)N(s)$ given by Equation 4.10 to obtain

$$\begin{aligned} \alpha'''(s) &= \frac{d}{ds}\big(k(s)N(s)\big) = k'(s)N(s) + k(s)N'(s) \\ &= k'(s)N(s) + k(s)[-\tau(s)B(s) - k(s)T(s)] \\ &= -[k(s)]^2 T(s) + k'(s)N(s) - k(s)\tau(s)B(s). \end{aligned}$$

Set

$$k_0 = k(0), \qquad k_0' = k'(0), \qquad \tau_0 = \tau(0). \tag{4.12}$$

Evaluating after differentiation gives

$$\begin{aligned} \alpha'(0) &= T_0, \\ \alpha''(0) &= k_0 N_0, \\ \alpha'''(0) &= -k_0^2 T_0 + k_0' N_0 - k_0 \tau_0 B_0. \end{aligned} \tag{4.13}$$

Taylor expansion with remainder. We write

$$\alpha(s) = \alpha(0) + s\alpha'(0) + \frac{s^2}{2}\alpha''(0) + \frac{s^3}{6}\alpha'''(0) + r_3(s),$$

with $\frac{\|r_3(s)\|}{|s|^3} \to 0$.

Remainder calculation. By integration by parts,

$$\begin{aligned} \frac{1}{2}\int_0^s (s-u)^2 \alpha'''(u)\,du &= \left[\frac{(s-u)^2}{2}\alpha''(u)\right]_0^s + \int_0^s (s-u)\alpha''(u)\,du \\ &= -\frac{s^2}{2}\alpha''(0) + \big[(s-u)\alpha'(u)\big]_0^s + \int_0^s \alpha'(u)\,du \\ &= -\frac{s^2}{2}\alpha''(0) - s\alpha'(0) + \alpha(s) - \alpha(0) \\ &= -\frac{s^2}{2}\alpha''(0) - s\alpha'(0) + \alpha(s) - p_0. \end{aligned} \tag{4.14}$$

Now,

$$r_3(s) = \frac{1}{6}\int_0^s (s-u)^3 \alpha^{(4)}(u)\,du. \tag{4.15}$$

Since

$$\frac{d}{du}\left[\frac{(s-u)^3}{6}\right] = -\frac{(s-u)^2}{2}, \tag{4.16}$$

integration by parts gives

$$\begin{aligned} r_3(s) &= \left[\frac{(s-u)^3}{6}\alpha'''(u)\right]_0^s + \frac{1}{2}\int_0^s (s-u)^2\alpha'''(u)\,du \\ &= -\frac{s^3}{6}\alpha'''(0) - \frac{s^2}{2}\alpha''(0) - s\alpha'(0) + \alpha(s) - p_0, \end{aligned}$$

since

$$\alpha(0) = p_0. \tag{4.17}$$

Thus,

$$\begin{aligned} \alpha(s) &= p_0 + s\alpha'(0) + \frac{s^2}{2}\alpha''(0) + \frac{s^3}{6}\alpha'''(0) + r_3(s), \\ \text{with } r_3(s) &= -\frac{s^3}{6}\alpha'''(0) - \frac{s^2}{2}\alpha''(0) - s\alpha'(0) + \alpha(s) - p_0. \end{aligned} \tag{4.18}$$

Remainder estimate. Recall $r_3(s) = \frac{1}{6}\int_0^s (s-u)^3\alpha^{(4)}(u)\,du$. Substitute $u = sv$ (so $u = 0$ corresponds to $v = 0$ and $u = s$ corresponds to $v = 1$). Then $r_3(s) = \frac{s^4}{6}\int_0^1 (1-v)^3\alpha^{(4)}(sv)\,dv$. Since α is smooth, take C_4 to be a bound for $\|\alpha^{(4)}\|$ on a small closed interval about zero. Now,

$$\begin{aligned} \|r_3(s)\| &= \frac{|s|^4}{6}\left\|\int_0^1 (1-v)^3\alpha^{(4)}(sv)\,dv\right\| \\ &\le \frac{|s|^4}{6}\int_0^1 (1-v)^3\|\alpha^{(4)}(sv)\|\,dv \\ &\le \frac{C_4|s|^4}{6}\int_0^1 (1-v)^3\,dv = \frac{C_4|s|^4}{24}. \end{aligned} \tag{4.19}$$

In particular,

$$\frac{\|r_3(s)\|}{|s|^3} \le \frac{C_4|s|}{24} \longrightarrow 0. \tag{4.20}$$

Now Equation 4.18 reads off as

$$\begin{aligned} \alpha(s) - p_0 &= s\alpha'(0) + \frac{s^2}{2}\alpha''(0) + \frac{s^3}{6}\alpha'''(0) + r_3(s) \\ &= sT_0 + \frac{k_0 s^2}{2}N_0 + \frac{s^3}{6}(-k_0^2 T_0 + k_0' N_0 - k_0\tau_0 B_0) + r_3(s). \end{aligned}$$

by applying Equation 4.13. Hence,

$$\alpha(s) - p_0 = \left(s - \frac{k_0^2 s^3}{6}\right)T_0 + \left(\frac{k_0 s^2}{2} + \frac{k_0' s^3}{6}\right)N_0 - \frac{k_0\tau_0 s^3}{6}B_0 + r_3(s). \tag{4.21}$$

Apply ρ on Equation 4.21 and then apply Equation 4.8:

$$\begin{aligned} \beta(s) = \rho(\alpha(s) - p_0) &= \left(s - \frac{k_0^2 s^3}{6}\right)\rho(T_0) + \left(\frac{k_0 s^2}{2} + \frac{k_0' s^3}{6}\right)\rho(N_0) \\ &\quad - \frac{k_0\tau_0 s^3}{6}\rho(B_0) + \rho(r_3(s)) \\ &= \left(s - \frac{k_0^2 s^3}{6}\right)e_1 + \left(\frac{k_0 s^2}{2} + \frac{k_0' s^3}{6}\right)e_2 - \frac{k_0\tau_0 s^3}{6}e_3 + \rho(r_3(s)). \end{aligned} \tag{4.22}$$

Define r_x, r_y, r_z to be the components of the rotated remainder:

$$\rho(r_3(s)) = r_x(s)e_1 + r_y(s)e_2 + r_z(s)e_3. \tag{4.23}$$

Then $r_x(s) = \langle \rho(r_3(s)), e_1 \rangle = \langle r_3(s), T_0 \rangle$, but $|r_x(s)| = |\langle r_3(s), T_0 \rangle| \leq \|\rho(r_3(s))\| = \|r_3(s)\|$. Similarly,

$$|r_y(s)| \leq \|r_3(s)\|, \qquad |r_z(s)| \leq \|r_3(s)\|. \tag{4.24}$$

Hence, we can rewrite Equation 4.22 as $\beta(s) = \big(s - \frac{k_0^2 s^3}{6} + r_x(s)\big)e_1 + \big(\frac{k_0 s^2}{2} + \frac{k_0' s^3}{6} + r_y(s)\big)e_2 + \big(-\frac{k_0 \tau_0 s^3}{6} + r_z(s)\big)e_3$. From Equation 4.6, one obtains the *canonical form*

$$\boxed{\begin{aligned} x(s) &= s - \frac{k_0^2}{6}s^3 + r_x(s), \\ y(s) &= \frac{k_0}{2}s^2 + \frac{k_0'}{6}s^3 + r_y(s), \\ z(s) &= -\frac{k_0\tau_0}{6}s^3 + r_z(s), \qquad |r_i(s)| \leq \frac{C_4|s|^4}{24} \quad (i = x, y, z). \end{aligned}} \tag{4.25}$$

Using the same remainders as in the plane calculation. Recall that the *tb*-plane is called the *rectifying plane* and the *tn*-plane is called the *osculating plane*. In the normalized coordinates (the frame being $\{e_1, e_2, e_3\}$), the rectifying plane is the e_1e_3-plane, i.e., the $y = 0$ plane, and the osculating plane is the e_1e_2-plane, i.e., the $z = 0$ plane. From Equation 4.25, one then obtains

$$\frac{y(s)}{s^2} = \frac{k_0}{2} + \frac{k_0'}{6}s + \frac{r_y(s)}{s^2} \longrightarrow \frac{k_0}{2} > 0, \qquad \frac{z(s)}{s^3} = -\frac{k_0\tau_0}{6} + \frac{r_z(s)}{s^3} \longrightarrow -\frac{k_0\tau_0}{6}. \tag{4.26}$$

Thus $y(s) > 0$ for all sufficiently small $s \neq 0$, on both sides of zero. If $\tau_0 \neq 0$, $z(s)$ changes sign at $s = 0$ because s^3 changes sign. In do Carmo's convention, negative torsion gives a crossing toward the positive binormal side, and positive torsion gives a crossing toward the negative binormal side.

In our normalized coordinates, the tangent line is the x-axis, i.e., e_1. The plane P_s contains $\beta(s) = (x(s), y(s), z(s))$ and $(x(s), 0, 0)$.

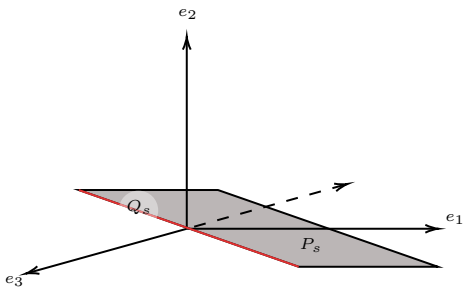


Diagram: two small oriented $\{e_1 = T, e_2 = N, e_3 = B\}$ coordinate sketches showing negative torsion crossing toward the positive binormal side and positive torsion crossing toward the negative binormal side, as on page 7 of the notes.

The line Q_s with direction $(0, y(s), z(s))$ lies in the P_s-plane. But since its x-coordinate is 0, the line lies in the YZ-plane; i.e., the plane P_s intersects the YZ-plane at the line Q_s. In fact, the P_s-plane is the one that contains the tangent vector e_1 at the origin and a nearby point $\beta(s)$ of the normalized curve. Now, $y(s) > 0$ for small $s \neq 0$. Dividing Q_s by $y(s)$ changes its length, not its direction. Thus,

$$P_s = \operatorname{span}\left\{e_1, \left(0, 1, \frac{z(s)}{y(s)}\right)\right\}. \tag{4.27}$$

So, looking along the x-axis, the cross-section of P_s is the line Q_s with slope

$$c(s) = \frac{z(s)}{y(s)}. \tag{4.28}$$

The canonical form gives $\frac{y(s)}{s^2} \to \frac{k_0}{2} > 0$ and $\frac{z(s)}{s^3} \to -\frac{k_0\tau_0}{6}$, as given by Equation 4.26. Therefore,

$$c(s) = \frac{z(s)}{y(s)} = s\,\frac{z(s)/s^3}{y(s)/s^2}. \tag{4.29}$$

From Equation 4.26, one obtains $\dfrac{z(s)/s^3}{y(s)/s^2} \to -\dfrac{\tau_0}{3}$ as $s \to 0$, so that $c(s) \to 0$ as $s \to 0$, from Equation 4.29. Hence, the second direction $(0, 1, c(s))$ in Equation 4.27 tends to e_2, while the first direction is always e_1. The limiting plane is therefore the XY-plane, $z = 0$, or the *osculating plane*.

In the next lecture, we will study global properties of plane curves.

References

[**Car76**] Manfredo P. do Carmo. *Differential Geometry of Curves and Surfaces*. Englewood Cliffs, NJ: Prentice-Hall, 1976. ISBN: 0-13-212589-7 (cited pp. 6, 10)